

RMC Homework - 2

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2.1) Given, ${}^A P$ vector rotated by θ about $\hat{Y}_A$ and by ϕ about $\hat{Z}_A$.
The Rotation matrix will be:

$${}^A R = {}^A_z R \cdot {}^A_y R \cdot P$$

Now, ${}^A_y R \rightarrow$ Rotation about Y_A will be given by:

$${}^A_y R = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

Similarly, ${}^A_z R$ (rotation about Z_A) will be:

$${}^A_z R = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Now, } {}^A R = {}^A_z R {}^A_y R \Rightarrow \begin{bmatrix} c\phi & -s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} c\theta c\phi & -s\phi & c\phi s\theta \\ s\phi c\theta & c\phi & s\phi s\theta \\ -s\theta & 0 & c\theta \end{bmatrix} \rightarrow {}^A R$$

Hence, the final Rotation Matrix looks like the above matrix.

$\downarrow$
 ${}^A R$

2.2) P is rotated by 45° about $\hat{Y}_A$ and 60° about $\hat{X}_A$.

$${}^A R = {}^A R_x {}^A R_y {}^A P$$

$${}^A_y R = \begin{bmatrix} \cos 45^\circ & 0 & \sin 45^\circ \\ 0 & 1 & 0 \\ -\sin 45^\circ & 0 & \cos 45^\circ \end{bmatrix} \Rightarrow \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$${}^A_x R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 60^\circ & -\sin 60^\circ \\ 0 & \sin 60^\circ & \cos 60^\circ \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 \\ 0 & \sqrt{3}/2 & 1/2 \end{bmatrix}$$

$${}^A R = {}^A_x R {}^A_y R {}^A P \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 \\ 0 & \sqrt{3}/2 & 1/2 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 3/2\sqrt{2} & 1/2 & -3/2\sqrt{2} \\ -1/2\sqrt{2} & \sqrt{3}/2 & 1/2\sqrt{2} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 1.0606 & 0.5 & -1.0606 \\ 0.3535 & 0.866 & -0.3535 \end{bmatrix}$$

This is the final Rotation matrix in the given order.

2.3)

Initially ${}^B P$ and ${}^A P$ are coincident. Now, ${}^B P$ is rotated about $\hat{x}_A$ by θ degrees and about $\hat{y}_A$ by ϕ degrees.

The resultant rotation matrix would be:

$${}^B_A R = {}^A_x R {}^y_y R {}^x_x R$$

$$R_y(\phi) = \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{bmatrix}$$

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$${}^B_A R = R_x(\theta) \cdot R_y(\phi) \Rightarrow \begin{bmatrix} c\phi & 0 & s\phi \\ s\phi s\theta & c\theta & -s\theta c\phi \\ -s\phi c\theta & s\theta & c\theta c\phi \end{bmatrix}$$

This is the final required Rotation matrix.

2.4)

$${}^B_A R = R_x(45) R_y(30)$$

$$R_y(30) = \begin{bmatrix} \cos 30 & 0 & \sin 30 \\ 0 & 1 & 0 \\ -\sin 30 & 0 & \cos 30 \end{bmatrix}; \quad R_x(45) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 45 & -\sin 45 \\ 0 & \sin 45 & \cos 45 \end{bmatrix}$$

Resultant B_R will be:

$$\begin{bmatrix} \cos 30 & 0 & \sin 30 \\ \sin 45 \sin 30 & \cos 45 & -\sin 45 \cos 30 \\ -\cos 45 \sin 30 & \sin 45 & \cos 45 \cos 30 \end{bmatrix} \Rightarrow \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.353 & 0.707 & -0.612 \\ -0.353 & 0.707 & 0.612 \end{bmatrix}$$

$\therefore$ This is the required Rotation Matrix.

2.12) Velocity is a 'free vector'. Hence, no effect due to translation.
The effect is only due to rotation and it is given by:

$$\begin{aligned} {}^A_V &= {}^A_B R \cdot {}^B_V \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 60 & -\sin 60 \\ 0 & \sin 60 & \cos 60 \end{bmatrix} \cdot \begin{bmatrix} 40 \\ 20 \\ 60 \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} 40 \\ 20(0.5) - 60(0.866) \\ 20(0.866) + 60(0.5) \end{bmatrix} \Rightarrow \begin{bmatrix} 40 \\ -41.96 \\ 47.32 \end{bmatrix} \\ &\quad \rightarrow {}^A_V. \end{aligned}$$

2.27) For ${}^A_B T$, translation of 3 units about $\hat{x}_B$ and rotation of 180° by $\hat{z}_B$.

$${}^A_B T = \begin{bmatrix} \cos 180 & -\sin 180 & 0 & 3 \\ \sin 180 & \cos 180 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.28)

for A_cT , $R_{xc}(-60) \rightarrow R_{yc}(-90) \rightarrow R_{zc}(-90)$
 followed by
 $T_{xA}(3) \rightarrow T_{zA}(2)$.

$$\begin{aligned} {}^A_cR &= R_{xc}(-60) \cdot R_{yc}(-90) \cdot R_{zc}(-90) \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.866 \\ 0 & -0.866 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.866 \\ 0 & -0.866 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & -1 \\ -0.5 & 0.866 & 0 \\ 0.866 & 0.5 & 0 \end{bmatrix} \end{aligned}$$

$$\therefore \text{Transformation Matrix} = \begin{bmatrix} 0 & 0 & -1 & 3 \\ -0.5 & 0.866 & 0 & 0 \\ 0.866 & 0.5 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.31) ${}^A_B T : R_{x_B}(90^\circ) \rightarrow R_{z_B}(180^\circ) \rightarrow T_{z_A}(2) \rightarrow T_{y_A}(4)$

$${}^A_B R = R_{x_B}(90^\circ) \cdot R_{y_B}(180^\circ)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$\therefore$ Transformation Matrix = $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 4 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

2.29) For ${}^B_C T : R_{x_C}(-60^\circ) \rightarrow R_{y_C}(-90^\circ) \rightarrow R_{z_C}(90^\circ) \rightarrow T_{z_B}(2)$

${}^B_C R$ will be: $R_{x_C}(-60^\circ) \cdot R_{y_C}(-90^\circ) \cdot R_{z_C}(90^\circ)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.866 \\ 0 & 0.866 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 0.5 & 0 & 0.866 \\ 0 & 1 & 0 \\ -0.866 & 0 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Final Transformation Matrix: $\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0.5 & 0.866 & 0 & 0 \\ -0.866 & 0.5 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$